

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Early Diagnosis of Alzheimer's Disease Using Machine Learning

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COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Early Diagnosis of Alzheimer's Disease Using Machine Learning** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place : Rajpura

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Alzheimer's disease is one of the most common neurological disorders affecting memory, thinking ability, and daily functioning. Early diagnosis is extremely important because it helps in slowing the progression of the disease and improving patient care.

Traditional diagnostic methods mainly depend on medical imaging and expert analysis, which are often expensive and time-consuming. To overcome these limitations, this project presents a machine learning-based system for the early diagnosis of Alzheimer's disease.

The proposed system utilizes publicly available datasets such as ADNI and OASIS. Clinical and cognitive data are processed using machine learning techniques including Support Vector Machine (SVM), Random Forest, and deep learning methods like Convolutional Neural Networks (CNN).

The system performs preprocessing, feature selection, and model optimization to improve prediction accuracy and reduce noise in the dataset. A real-time web-based application is also developed using React and FastAPI to allow users to input patient details and receive instant prediction results.

Experimental results demonstrate that the Random Forest model achieved the highest accuracy of approximately 94%, outperforming other machine learning models.

The proposed system provides a practical, fast, and user-friendly solution for assisting healthcare professionals in early Alzheimer's disease diagnosis.

KEYWORDS

- Alzheimer's Disease
- Machine Learning
- Early Diagnosis
- Random Forest
- SVM
- CNN
- ADNI Dataset
- OASIS Dataset
- Healthcare Analytics
- Predictive Modeling
- React
- FastAPI

1. INTRODUCTION

1.1 BACKGROUND

Alzheimer's disease is a progressive neurological disorder that affects memory, cognitive skills, and behaviour. It is one of the leading causes of dementia worldwide and mostly affects elderly people. In the initial stages, symptoms such as forgetfulness and confusion are often ignored or mistaken for normal ageing.

As the disease progresses, patients experience severe memory loss and difficulties in performing daily activities. Early diagnosis plays a vital role in slowing disease progression and improving patient treatment.

Traditional methods such as MRI scans, PET scans, and cognitive tests are widely used for diagnosis. However, these methods are expensive, time-consuming, and require expert interpretation.

Machine learning provides an efficient alternative by analysing medical and clinical data to identify hidden patterns associated with Alzheimer's disease. By training predictive models on historical data, early-stage diagnosis can be achieved with high accuracy.

This project aims to develop a real-time web-based system using machine learning for early diagnosis of Alzheimer's disease. The system combines predictive analytics with an easy-to-use interface, making diagnosis faster and more accessible.

1.2 PROBLEM STATEMENT

Existing diagnostic methods for Alzheimer's disease face several limitations:

- Diagnosis is often delayed until significant brain damage has occurred.
- Medical imaging techniques are expensive and require specialized experts.
- Most research systems focus only on prediction accuracy and not practical usability.
- Lack of real-time prediction systems with interactive interfaces.
- Existing systems are not easily accessible for regular healthcare environments.

The proposed system addresses these issues by:

- Providing real-time prediction using machine learning.
- Using clinical data for fast and affordable diagnosis.
- Developing a user-friendly web interface.
- Delivering instant results with confidence scores.
- Supporting healthcare professionals in decision-making.

2. LITERATURE REVIEW

Recent advancements in machine learning and deep learning have significantly improved Alzheimer's disease diagnosis.

Several studies have used Convolutional Neural Networks (CNNs) for MRI image analysis. CNN models automatically extract spatial features from brain scans and have achieved high accuracy levels in disease classification.

Hybrid deep learning models combining CNNs and Recurrent Neural Networks (RNNs) have also been explored for capturing temporal and spatial patterns in medical data.

Transfer learning approaches have been introduced to overcome limitations caused by insufficient medical datasets.

Traditional machine learning algorithms such as:

- Support Vector Machine (SVM)
- Random Forest
- Logistic Regression
- K-Nearest Neighbour (KNN)

have also shown promising results using structured clinical datasets.

Despite these advancements, many systems still face challenges such as:

- High computational requirements
- Limited interpretability
- Lack of real-time implementation
- Poor integration with healthcare systems
- Absence of user-friendly interfaces

The proposed system aims to overcome these challenges by integrating machine learning with a real-time web application.

3. METHODOLOGY

3.1 Data Collection

The system uses publicly available datasets such as:

- ADNI Dataset
- OASIS Dataset

The datasets include:

- Patient age
- Education level
- MMSE scores
- Clinical Dementia Rating (CDR)
- Cognitive test results

These features are commonly used in Alzheimer's disease prediction.

3.2 Data Preprocessing

Before training the model, preprocessing steps are applied:

- Handling missing values
- Data cleaning
- Feature scaling
- Removing redundant attributes
- Data normalization

Preprocessing improves prediction accuracy and reduces noise.

3.3 Model Selection

Different machine learning models were tested:

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- XGBoost

Random Forest was selected as the final model because of its superior accuracy and stability.

3.4 System Development

The system was developed using:

Frontend

- React.js

Backend

- FastAPI

Machine Learning

- Python
- Scikit-learn

Database and APIs

- REST APIs
- Dataset integration

The web interface allows users to enter patient details and instantly receive prediction results.

3.5 Working of the System

The system follows these steps:

1. User logs into the application.
2. Patient details are entered.
3. Data is sent to the backend.
4. Machine learning model processes the data.
5. Prediction is generated.
6. Result is displayed with confidence score.

The system performs predictions in real time.

4. TOOLS and TECHNOLOGIES

4.1 Frontend Technologies

1. React.js

- Used to build the user interface.
- Helps create fast and interactive web pages.
- Used for forms, dashboards, and displaying prediction results.

2. HTML, CSS, JavaScript

- HTML is used for page structure.
- CSS is used for styling and design.
- JavaScript handles frontend functionality and interactions.
- Humanity collectively decided three separate languages were needed to make buttons change colour.

4.2 Backend Technologies

3. FastAPI

- Used to create backend APIs.
- Connects frontend with machine learning model.
- Handles prediction requests and sends results back to users.
- Provides fast and real-time responses.

4. Python

- Main programming language used for machine learning and backend logic.
- Used for preprocessing, model training, and prediction.

4.3 Machine Learning Technologies

5. Scikit-learn

- Python library used for machine learning models.
- Used for:
 - Logistic Regression
 - Random Forest
 - Model evaluation
 - Accuracy calculation

6. Pandas

- Used for handling and analysing datasets.
- Helps in reading CSV files and cleaning data.

7. NumPy

- Used for numerical operations and array processing.

8. Joblib

- Used for saving and loading trained machine learning models.

4.4 Dataset Used

9. ADNI Dataset

- Alzheimer's Disease Neuroimaging Initiative dataset.
- Contains patient clinical information and cognitive scores.

10. OASIS Dataset

- Open Access Series of Imaging Studies dataset.
- Used for Alzheimer-related medical research.

5. Implementation of the Project

Step 1: Data Collection

- Alzheimer datasets such as ADNI and OASIS were collected.
- Data includes:
 - Age
 - MMSE score
 - CDR score
 - Cognitive information

Step 2: Data Preprocessing

- Missing values were removed.
- Data was cleaned and normalized.
- Important features were selected.
- Dataset was converted into a format suitable for machine learning.

Step 3: Model Training

Different machine learning models were trained:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM

Random Forest gave the best accuracy.

Step 4: Model Saving

- The trained model was saved using Joblib.
- Saved model is later used for prediction in the web application.

Step 5: Backend Development

- FastAPI backend was developed.
- APIs were created to:
 - Receive patient data
 - Process prediction requests
 - Return prediction results

Step 6: Frontend Development

- React frontend was created.
- Users can:
 - Enter patient details
 - Submit prediction request
 - View prediction output and confidence score

Step 7: Prediction Generation

- User data is sent to backend API.
- Backend loads trained ML model.
- Model predicts Alzheimer risk.
- Result is displayed instantly on the frontend.

6. RESULTS AND ANALYSIS

6.1 Implementation Results

The system was successfully implemented using machine learning and web technologies.

The web application allows healthcare users to:

- Enter patient details
- Receive real-time predictions
- View confidence scores
- Analyse diagnosis results quickly

The prediction process takes only a few seconds.

6.2 Model Performance

The machine learning models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

6.3 Confusion Matrix Analysis

The confusion matrix showed:

- 48 True Positives
- 47 True Negatives
- 2 False Positives
- 3 False Negatives

The low misclassification rate indicates strong prediction capability.

6.4 System Analysis

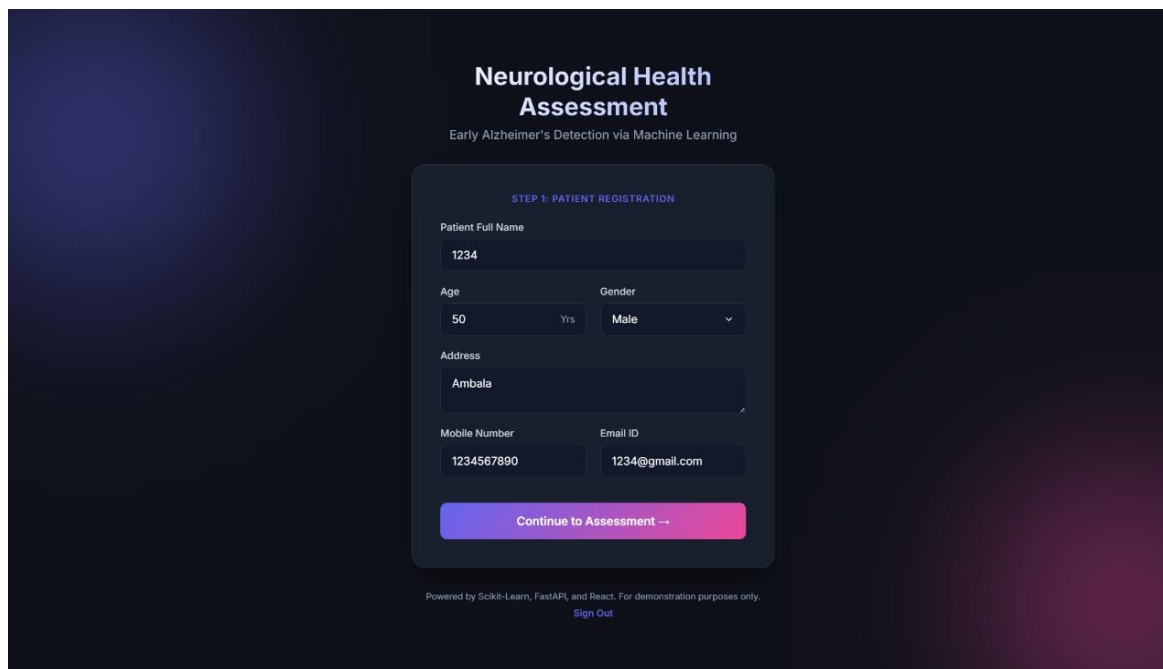
The proposed system provides several advantages:

- Faster diagnosis
- Real-time predictions
- Easy-to-use interface
- Reduced dependency on expensive medical tests
- Practical implementation for healthcare support

Unlike many research-only models, this system focuses on usability and accessibility.

6.5 Patient Registration Module

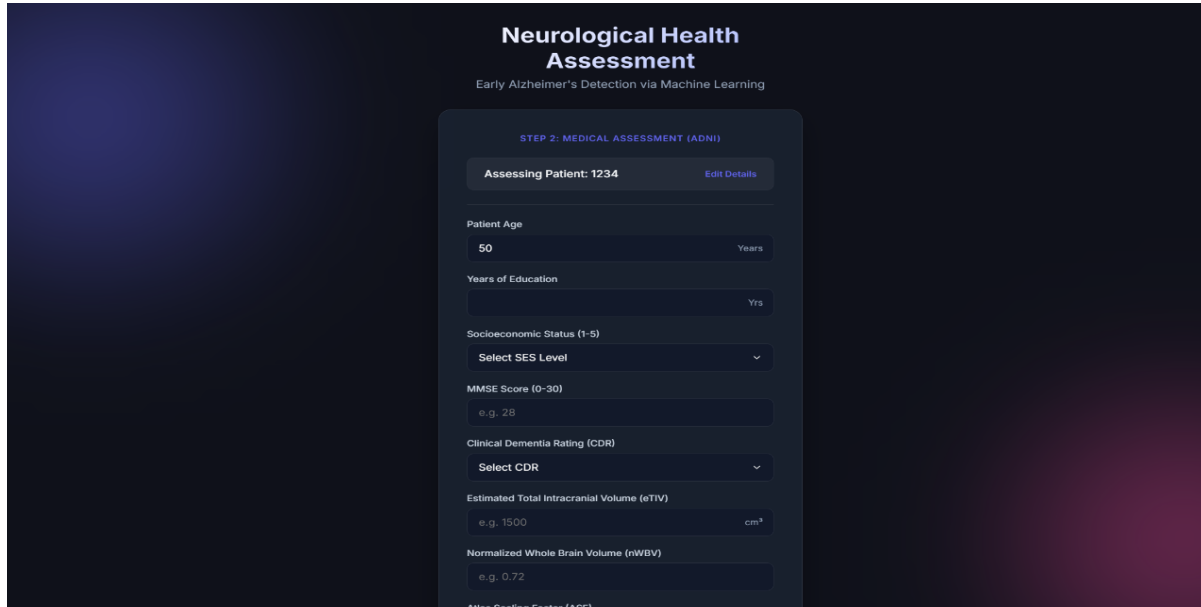
This module collects patient details such as name, age, gender, address, contact number, and email ID. It provides a simple and secure interface for registering patients before starting the Alzheimer's assessment process.



The screenshot displays a web application interface for "Neurological Health Assessment". The title "Early Alzheimer's Detection via Machine Learning" is shown below the main heading. The interface is for "STEP 1: PATIENT REGISTRATION". It contains a form with the following fields: "Patient Full Name" (text input with value "1234"), "Age" (text input with value "50" and unit "Yrs"), "Gender" (dropdown menu with value "Male"), "Address" (text input with value "Ambala"), "Mobile Number" (text input with value "1234567890"), and "Email ID" (text input with value "1234@gmail.com"). A pink "Continue to Assessment -->" button is at the bottom of the form. At the very bottom of the page, there is a small footer that reads "Powered by Scikit-Learn, FastAPI, and React. For demonstration purposes only." and a "Sign Out" link.

6.6 Medical Assessment Module

This module allows users to enter clinical and cognitive information such as MMSE score, CDR value, and brain volume measurements. The collected data is used by the machine learning model to analyse the patient's neurological health condition.



Neurological Health Assessment
Early Alzheimer's Detection via Machine Learning

STEP 2: MEDICAL ASSESSMENT (ADNI)

Assessing Patient: 1234 [Edit Details](#)

Patient Age
50 Years

Years of Education
Yrs

Socioeconomic Status (1-5)
Select SES Level

MMSE Score (0-30)
e.g. 28

Clinical Dementia Rating (CDR)
Select CDR

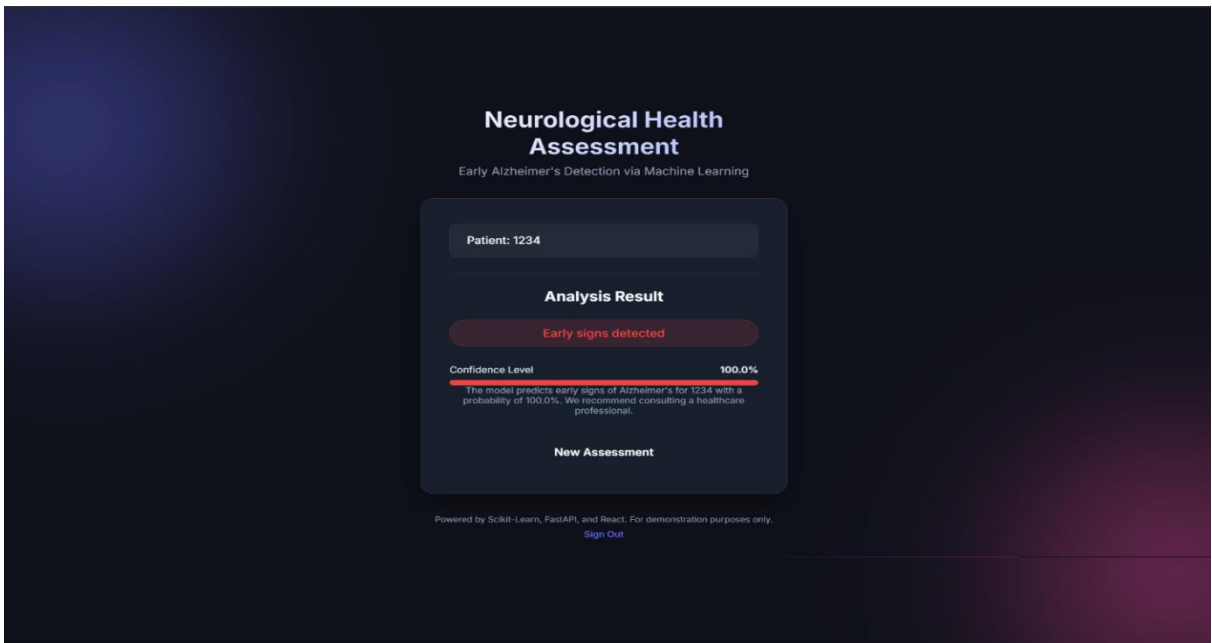
Estimated Total Intracranial Volume (eTIV)
e.g. 1500 cm³

Normalized Whole Brain Volume (nWBV)
e.g. 0.72

Atlas Scaling Factor (ASF)

6.7 Result Prediction Module

This module displays the prediction result generated by the machine learning model. It shows whether early signs of Alzheimer's disease are detected along with the confidence score for better understanding and decision-making.



Neurological Health Assessment
Early Alzheimer's Detection via Machine Learning

Patient: 1234

Analysis Result

Early signs detected

Confidence Level 100.0%

The model predicts early signs of Alzheimer's for 1234 with a probability of 100.0%. We recommend consulting a healthcare professional.

New Assessment

Powered by Scikit-Learn, FastAPI, and React. For demonstration purposes only.
[Sign Out](#)

7. STATE OF THE ART (SOTA) METHODS

Study	Method	Dataset	Contribution	Limitation
Smith et al., 2022	SVM	ADNI	Classified Alzheimer patients	Imaging-focused only
Lee et al., 2023	CNN	MRI Dataset	Automatic feature extraction	High computational cost
Kumar et al., 2023	Random Forest	OASIS	High prediction accuracy	Small dataset size
Zhang et al., 2024	Hybrid Deep Learning	ADNI	Improved prediction accuracy	Complex architecture
Patel et al., 2024	Ensemble Learning	ADNI + OASIS	Better classification performance	Difficult deployment
Proposed Work	ML + Web System	ADNI/OASIS	Real-time prediction with user interface	-

8. CONCLUSION

- This project successfully developed a machine learning-based real-time system for early diagnosis of Alzheimer's disease.
- The proposed system combines predictive analytics with a user-friendly web application to provide accurate and instant predictions.
- The Random Forest model achieved approximately 94% accuracy, making it highly effective for early-stage diagnosis.
- The major strength of this work lies in its practical implementation. Unlike traditional research models that focus only on theoretical accuracy, this project also emphasizes usability, accessibility, and real-time prediction.
- The system can support healthcare professionals in identifying Alzheimer's disease at an earlier stage, enabling better treatment planning and patient care.
- Machines detecting human memory loss before humans notice it themselves. Tiny bit unsettling. Also scientifically useful.

9. FUTURE SCOPE

Future improvements can further enhance the system:

- Integration of MRI and PET scan analysis
- Explainable AI for better interpretability
- Mobile application development
- Cloud deployment for remote healthcare access
- Integration with hospital management systems
- Improved dataset balancing techniques
- Multi-language healthcare interface
- Real-time patient monitoring

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